



Khansa Rasheed

Biomedical Engineering Student

UET Lahore, Narowal Campus · 4th Sem

CONTACT

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📁 Portfolio: khansarasheed.github.io

📍 Bhera, Sargodha, Punjab

ACADEMIC SNAPSHOT

BS Biomedical Eng. (2024–2028)

UET Lahore, Narowal Campus

FSc Pre-Medical (2022–2024)

Punjab College Bhera (1044/1200)

Matric (2020–2022)

Govt. High School Bhera (1011/1100)

TECHNICAL SKILLS

Arduino / Embedded Systems **82%**

Circuit Design & Analysis **78%**

Bio-Signal Processing **75%**

Digital Logic & Gates **80%**

Python (Fundamentals) **60%**

PCB & Sensor Integration **70%**

LANGUAGES

Urdu & English (Professional) **95%**

Punjabi (Fluent) **90%**

Khansa Rasheed

Biomedical Engineering Intern | UET Lahore, Narowal Campus | 4th Semester

A dedicated Biomedical Engineering student with hands-on experience in embedded systems, analog circuit design, and bio-signal acquisition. Passionate about bridging cutting-edge engineering with life-saving healthcare technology. Seeking an internship to apply academic knowledge in a real-world clinical or R&D environment.

Education

BS Biomedical Engineering

University of Engineering & Technology (UET) Lahore, Narowal Campus

📅 2024 – 2028

📍 Narowal, Punjab, Pakistan

FSc Pre-Medical

Punjab College Bhera | Marks: 1044 / 1200

📅 2022 – 2024

📍 Bhera, Sargodha, Punjab

Secondary School Certificate (Matric)

Govt. High School Bhera | Marks: 1011 / 1100

📅 2020 – 2022

📍 Bhera, Sargodha, Punjab

Engineering Projects

🔧 Arduino-Based 4-Way Traffic Signal Control System

Designed and implemented a fully automated traffic management system using Arduino UNO. Programmed adaptive timing logic to control variable traffic density across four lanes using embedded C. Demonstrated live with integrated LEDs and breadboard hardware.

🔧 Temperature Sensor Using Differential Amplifier

Constructed an analog temperature measurement circuit using an op-amp differential amplifier paired with a thermistor bridge. Calibrated output voltage for high-sensitivity temperature readings. Full hardware demonstration published on YouTube.

📺 [Watch Demo on YouTube](#)

🔧 Intelligent ICU Patient Monitoring System (Logic Gates)

Designed a combinational logic circuit simulating a multi-parameter ICU alert system. Encoded threshold conditions for heart rate, SpO₂, and temperature using NAND/NOR/AND gates — triggering alarms without a microcontroller.

🔧 ECG Signal Acquisition and Conditioning

Developed a complete ECG front-end: instrumentation amplifier, 0.5–150 Hz band-pass filter, and 60 Hz notch filter. Captured and conditioned realistic cardiac waveforms demonstrating core biopotential measurement principles.

Core Competencies & Tools

Engineering Skills

✔️ Arduino / AVR Microcontrollers

✔️ Bio-Signal Processing (ECG/EMG)

✔️ Embedded C Programming

✔️ Technical Report Writing

✔️ Analog & Digital Circuit Design

✔️ Op-Amp & Instrumentation Amp

✔️ PCB & Sensor Integration

✔️ Hardware Troubleshooting

Software & Simulation Tools

🛠️ Proteus

📡 LTSPICEXVII

📄 Origin

🔬 TINA_tj

🔗 VS Code

Achievements & Highlights

Consistently ranked among top performers in Matric and FSc at respective institutions.

Completed **4 rigorous semesters** of Biomedical Engineering at UET Lahore, Narowal Campus while simultaneously executing multiple hands-on hardware projects.

Documented and published a hardware project demo on YouTube — demonstrating professional